A Zero-Knowledge Decentralized Identity Protocol for Privacy-Preserving Mustahik Verification in Blockchain-Based Zakat Management Systems

Research Proposal

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Research Report

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Abstract

Zakat, one of the five pillars of Islam, constitutes a mandatory system of wealth redistribution with profound social welfare implications. Despite the increasing digitization of zakat management, contemporary systems remain burdened by critical deficiencies: opacity in fund disbursement, centralized control susceptible to corruption and mismanagement, and, most critically, the exposure of mustahik (zakat recipients) personal and socioeconomic data during eligibility verification. This research proposes a novel Zero-Knowledge Decentralized Identity (ZK-DID) protocol specifically engineered for privacy-preserving mustahik verification within blockchain-based zakat management systems.

The proposed architecture integrates zero-knowledge proof systems, specifically zk-SNARKs (Zero-Knowledge Succinct Non-Interactive Arguments of Knowledge), with decentralized identifier standards to enable cryptographic verification of a recipient's eligibility without disclosing underlying personal attributes or financial records. The protocol is designed to satisfy simultaneous requirements of privacy, verifiability, scalability, and Sharia compliance. Experimental evaluation focuses on proof generation overhead, on-chain verification costs, and resistance to common identity-linkage attacks. This work contributes a domain-specific identity layer that addresses both the technical and ethical imperatives of privacy in Islamic social finance infrastructure.

Keywords: Zero-Knowledge Proofs, zk-SNARKs, Decentralized Identity, Blockchain, Zakat Management, Mustahik Verification, Privacy-Preserving Protocols, Islamic FinTech, Self-Sovereign Identity

I. Introduction

1.1 Research Background

Zakat management in the contemporary landscape stands at the intersection of centuries-old Islamic jurisprudence and modern digital infrastructure demands. Globally, zakat collection is estimated to reach between USD 200 billion and USD 1 trillion annually, yet a significant portion fails to reach legitimate mustahik due to systemic inefficiencies, misappropriation, and institutional distrust (World Bank, 2020). The opacity inherent in centralized zakat institutions has eroded public confidence, while manual verification processes for mustahik eligibility remain vulnerable to fraud and administrative bottlenecks.

The advent of distributed ledger technology has prompted research into blockchain-based zakat systems that offer immutability, transparency, and automated disbursement via smart contracts. Pioneering works such as those by Mustafa et al. (2021) and Hassan et al. (2020) have demonstrated that Ethereum-based zakat platforms can significantly reduce intermediary overhead and improve auditability. However, these systems introduce a paradox: by placing transaction records on a public ledger, they inadvertently expose the socioeconomic status and personal circumstances of mustahik to public scrutiny. The stigma associated with receiving charity in Muslim-majority societies means that such exposure can have serious psychosocial and economic consequences for recipients.

Zero-Knowledge Proofs (ZKPs), formalized by Goldwasser, Micali, and Rackoff (1985), offer a compelling cryptographic resolution. A ZKP allows a prover to convince a verifier that a statement is true without revealing any information beyond the truth of that statement itself. In the context of mustahik verification, a ZKP-based system could allow a recipient to prove they satisfy eligibility criteria without disclosing the actual values of those attributes. Recent advances in succinct non-interactive ZKP constructions, particularly Groth16, PLONK, and their variants, have made on-chain proof verification computationally feasible (Groth, 2016; Gabizon et al., 2019).

Decentralized Identity (DID) standards, as defined by the W3C DID Core specification (Sporny et al., 2022), further complement this approach by enabling self-sovereign identity management in which individuals control their own credentials without dependence on a central authority. Combining ZKPs with DID frameworks creates a powerful identity layer capable of selective disclosure. This paradigm aligns with the maqasid al-shariah (objectives of Islamic law), specifically the protection of nafs (life and dignity) and mal (wealth) of individuals. Despite the theoretical promise of this convergence, no prior work has proposed a domain-specific ZK-DID protocol tailored to the unique requirements of zakat management.

1.2 Research Objectives

The primary objective of this research is to design, implement, and evaluate a Zero-Knowledge Decentralized Identity protocol for privacy-preserving mustahik verification in blockchain-based zakat management systems. The specific objectives are:

1. To formalize the privacy requirements for mustahik verification as cryptographic predicates expressible within a zk-SNARK circuit framework.
2. To design a ZK-DID protocol architecture integrating W3C Decentralized Identifier standards with zk-SNARK-based credential verification in a blockchain-native zakat management context.
3. To implement a prototype system on an EVM-compatible blockchain, including smart contracts for on-chain proof verification and DID registry management.

4. To evaluate the protocol with respect to computational overhead of proof generation, on-chain verification gas costs, and privacy guarantees against identity-linkage and re-identification attacks.
5. To assess the practical feasibility of deployment within existing zakat institution (amil) operational frameworks, including Sharia compliance considerations.

1.3 Research Questions

This study addresses the following research questions:

1. How can zero-knowledge proof systems be formally applied to encode Sharia-compliant mustahik eligibility criteria as verifiable predicates without exposing underlying personal or financial attributes?
2. What architectural integration is required to couple W3C Decentralized Identifier standards with zk-SNARK-based credential verification in a blockchain-native zakat management context?
3. What are the practical computational and economic trade-offs -- proof generation time, circuit complexity, and on-chain gas consumption -- of the proposed ZK-DID protocol at scale?
4. To what extent does the proposed protocol satisfy both formal privacy properties (zero-knowledge, soundness, completeness) and the operational requirements of amil institutions?

1.4 Benefit of Research

This research offers contributions of significance to multiple stakeholder groups and knowledge domains:

- Academic Contribution: Extends the body of applied cryptography literature by producing a domain-specific ZKP circuit design for socioeconomic eligibility verification. The proposed protocol will contribute an open-source implementation accessible to blockchain developers and Islamic FinTech practitioners.
- Social and Ethical Impact: Directly addresses the dignity and privacy rights of zakat recipients, a population that is often economically vulnerable. By ensuring that the act of seeking zakat does not require disclosure of sensitive personal circumstances, the protocol reduces barriers to legitimate access.
- Policy and Institutional Value: Provides a technically grounded framework for modernizing eligibility verification workflows in a manner that is both digitally auditable and privacy-respecting. This can serve as a reference architecture for national zakat bodies in Indonesia, Malaysia, and other Muslim-majority jurisdictions.
- Technological Advancement: Demonstrates how deep representation learning and ZKP circuit design can uncover latent eligibility verification patterns in unlabeled credential data, improving interpretability and scalability in Islamic FinTech analytics.

II. Literature Review

2.1 Blockchain-Based Zakat Management Systems

The intersection of blockchain technology and Islamic social finance has attracted increasing scholarly attention since 2017. Mustafa et al. (2021) proposed a foundational Ethereum-based zakat distribution framework employing ERC-20 tokens to represent zakat funds, demonstrating that smart contract automation could eliminate intermediary delays and provide an immutable audit trail. Their system reduced disbursement processing time by approximately 60% in a simulated institutional environment. However, the system inherited all transparency properties of the underlying public blockchain, inadvertently exposing recipient addresses to potential de-anonymization through transaction graph analysis.

Hassan et al. (2020) extended this work by incorporating a multi-signature governance mechanism, enabling collective decision-making among amil committee members for high-value disbursements. Subsequent work by Rahman and Zakaria (2023) explored the use of Hyperledger Fabric as an alternative model, arguing that permissioned ledgers provide de facto privacy through access control. This approach, however, introduces centralization trade-offs and undermines the trustless verification that constitutes a core value proposition of public blockchain infrastructure. The literature thus reveals a persistent tension: transparency, necessary for institutional accountability, conflicts directly with the privacy needs of recipients.

2.2 Zero-Knowledge Proofs: Theoretical Foundations and Applied Systems

The theoretical foundations of zero-knowledge proofs were established by Goldwasser, Micali, and Rackoff (1985) in their seminal work defining the three properties of completeness, soundness, and zero-knowledge. Groth (2016) introduced the Groth16 construction, a pairing-based zk-SNARK requiring a circuit-specific trusted setup, which remains the most widely deployed proof system due to its highly optimized verification time and proof size of approximately 192 bytes. Groth16 underlies the privacy mechanisms of Zcash and numerous layer-2 ZK rollup architectures.

The Universal PLONK construction by Gabizon, Williamson, and Ciobanu (2019) introduced a universal and updateable structured reference string (SRS), permitting a single trusted setup ceremony to be reused across multiple circuits. This property substantially reduces the ceremony overhead in multi-circuit deployments, a relevant consideration for a zakat system encoding multiple distinct eligibility predicates. Subsequent variants such as TurboPlonk and UltraPlonk extended the arithmetic capabilities of the constraint system to support custom gates and lookup arguments, enabling more efficient encoding of range proofs and set membership checks that are central to income and asset verification.

2.3 Decentralized Identity Standards and Self-Sovereign Identity

The W3C Decentralized Identifiers (DID) Core specification (Sporny et al., 2022) defines a global standard for decentralized, cryptographically verifiable identifiers that are independent of any central registry or authority. The W3C Verifiable Credentials (VC) Data Model (Sporny et al., 2019) complements the DID specification by providing a standard envelope for cryptographically signed attribute assertions issued by credential issuers. In the proposed system, a government or certified welfare assessment body would act as the credential issuer, attesting to a mustahik's eligibility criteria in the form of a VC.

Self-Sovereign Identity (SSI) architectures have been extensively examined in the literature. Muhle et al. (2018) provide a comprehensive survey of SSI architectures, identifying the three-party trust triangle (issuer-holder-verifier) as the foundational model. The BBS+ signature scheme (Boneh, Boyen, and Shacham, 2004) has emerged as a preferred primitive for ZK-compatible credential systems due to its support for efficient proofs of knowledge of signed messages, and has been adopted in several W3C Verifiable Credential implementations.

2.4 Privacy-Preserving Credential Verification: Recent Advances

The combination of ZKPs with verifiable credentials has been explored in several recent works. Bichsel et al. (2010) introduced the concept of the Privacy-ABC framework, later formalized in the ABC4Trust project (Camenisch et al., 2012), which provided a common API for interoperable anonymous credential systems. More recently, Pauwels (2021) proposed zkKYC, a zero-knowledge Know-Your-Customer framework applicable to regulated financial environments, demonstrating that ZK-based compliance verification is achievable within existing AML/CFT regulatory frameworks. Their work is directly analogous to the zakat eligibility use case, though without the domain-specific Sharia compliance requirements.

The use of range proofs -- ZKPs that demonstrate a committed value lies within a specified interval without revealing the value -- is central to income and asset verification. Bulletproofs (Bunz et al., 2018) provide logarithmically sized range proofs without a trusted setup, making them attractive for unbounded range assertions. However, their verification cost is linear in the number of proofs, which may present scalability challenges for high-throughput zakat disbursement cycles.

Table 1. Summary of Related Literature

Source	**Domain**	**Contribution**	**Limitation**
Mustafa et al. (2021)	Blockchain Zakat	Ethereum ERC-20 zakat framework; 60% reduction in disbursement time.	No privacy mechanisms; recipient addresses linkable on-chain.
Hassan et al. (2020)	Blockchain Zakat	Multi-signature governance for zakat disbursement accountability.	Pseudonymous addressing; no identity protection.
Rahman & Zakaria (2023)	Blockchain Zakat	Hyperledger Fabric permissioned ledger for de-facto privacy.	Centralization trade-offs; not trustless.
Groth (2016)	ZKP Systems	Groth16 zk-SNARK; ~192 byte proof; O(1) pairing verification.	Circuit-specific trusted setup required.
Gabizon et al. (2019)	ZKP Systems	PLONK universal SRS reusable across multiple circuits.	Higher proof generation overhead than Groth16.
Pauwels (2021)	Privacy Identity	zkKYC ZK framework for financial compliance verification.	No Sharia-specific eligibility predicates.

Sporny et al. (2022)	Decentralized ID	W3C DID Core standard; decentralized cryptographic identifiers.	No ZKP integration defined in standard.

2.5 Research Gap

Despite substantial literature addressing blockchain-based zakat management, zero-knowledge proof systems, decentralized identity standards, and privacy-preserving credential verification as independent research domains, a critical convergence gap remains unaddressed. No prior work has proposed a domain-specific ZK-DID protocol engineered to satisfy the simultaneous technical, ethical, and jurisprudential requirements of mustahik verification within blockchain-based zakat management systems.

Three interconnected gaps can be identified. First, while blockchain-based zakat systems have demonstrated operational feasibility, none have integrated privacy-preserving identity mechanisms beyond pseudonymous addressing. Second, existing ZKP-based credential frameworks have been designed for generalized financial compliance contexts and do not account for the domain-specific eligibility predicates defined under Islamic jurisprudence -- nisab thresholds, hawl assessment periods, and the classification of recipients across the eight asnaf categories. Third, the integration of W3C DID standards with ZKP-based credential verification has not been explored within the Islamic social finance domain. This research directly addresses all three gaps.

III. Research Methodology

3.1 Research Design

This research adopts a Design Science Research (DSR) methodology (Hevner et al., 2004), which is well-suited to artifact-oriented contributions in the information systems and applied cryptography domains. The DSR cycle encompasses problem identification, artifact design, prototype construction, and rigorous evaluation against defined requirements. The research is structured in four sequential phases: (1) Formal Specification, (2) Circuit and Protocol Design, (3) Prototype Implementation, and (4) Evaluation and Analysis.

3.2 Phase 1: Formal Specification of Eligibility Predicates

The initial phase involves the formal specification of mustahik eligibility criteria as verifiable computational predicates. In collaboration with Islamic jurisprudence scholars and qualified zakat practitioners, the research will catalogue the complete set of eligibility conditions across the eight asnaf categories, focusing initially on the fakir (destitute) and miskin (poor) categories which represent the largest recipient populations. Each condition will be expressed as a Boolean predicate over committed attribute values:

> *Let A = {a1, a2, ..., an} be the set of private attribute values held by a mustahik claimant (e.g., monthly income, total non-exempt assets, number of dependents). Let T = {t1, t2, ..., tn} be the publicly known threshold values for the current zakat cycle. The eligibility predicate P is defined as a Boolean circuit C such that C(A, T) = 1 if and only if the claimant satisfies all required conditions. The zero-knowledge statement to be proven is: 'I know private attributes A such that C(A, T) = 1 and A is committed to in commitment Com(A)'.*

This formalization will be documented as a formal predicate specification using R1CS (Rank-1 Constraint System) notation, the standard intermediate representation for zk-SNARK circuits.

3.3 Phase 2: ZK-DID Protocol Architecture Design

The second phase designs the complete protocol architecture. The proposed ZK-DID protocol for zakat involves five principal actors: the Government Data Authority (credential issuer), the Mustahik Claimant (credential holder and ZKP prover), the Amil Institution (verifier and on-chain transaction initiator), the Blockchain Network (verification and disbursement execution), and the DID Registry (on-chain identifier resolution). The ull architecture is illustrated in Figure 1 below.

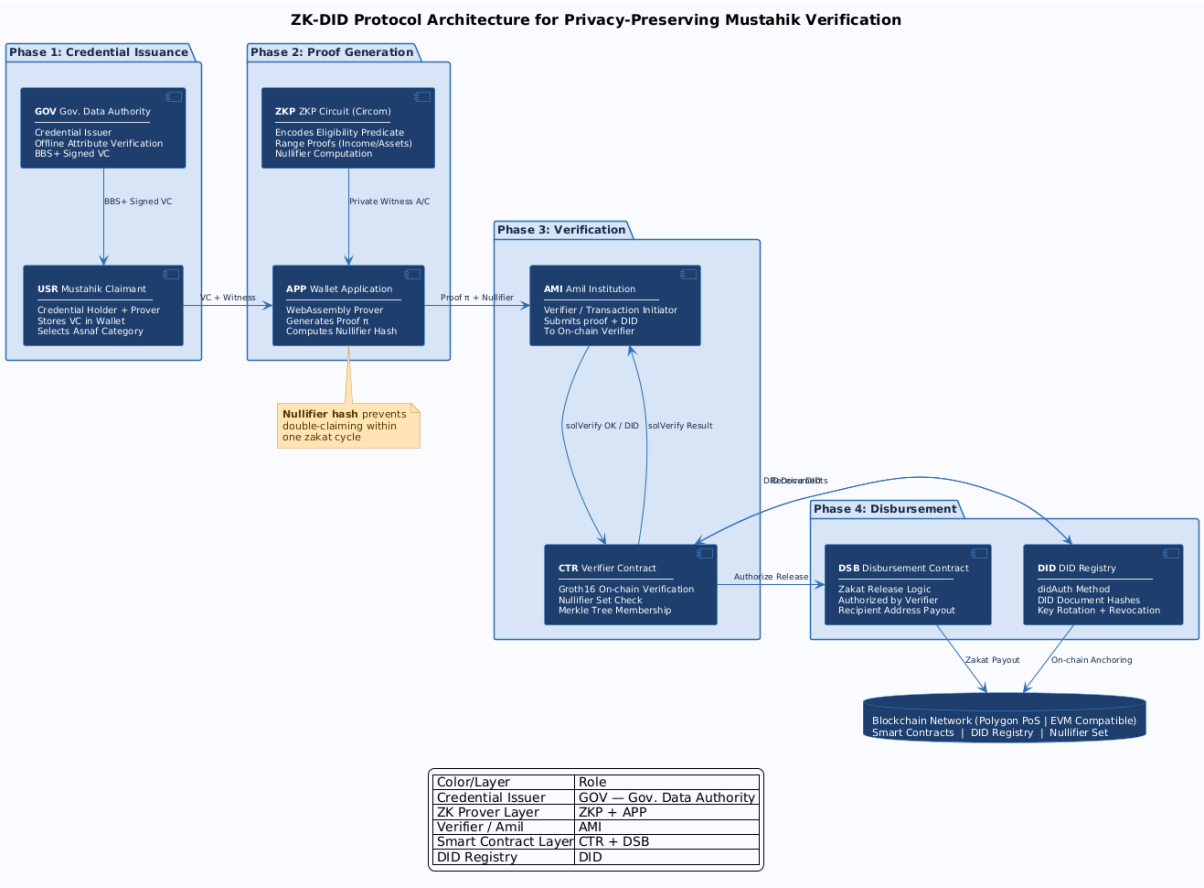

Figure 1. ZK-DID Protocol Architecture -- Five Principal Actors and Protocol Phase Flow

Table 2. ZK-DID Protocol Flow Across Principal Actors

Phase	**Actor**	**Description**
1. Credential Issuance	Government Data Authority	Conducts offline attribute verification and issues a BBS+-signed Verifiable Credential (VC) over the attribute vector A conforming to the W3C VC Data Model. The VC is stored in the claimant's personal wallet application.
2. Proof Generation	Mustahik Claimant	Selects the applicable asnaf category and invokes the ZKP circuit C. Uses VC attributes as the private witness to generate a zk-SNARK proof pi, together with a nullifier hash to prevent double-claiming within a single zakat cycle.
3. Verification	Amil Institution	Submits the proof pi, nullifier hash, and claimant's DID to the on-chain Verifier smart contract. The contract verifies pi against the pre-compiled verification key and checks the nullifier against the on-chain nullifier set.
4. Disbursement	Blockchain Network	Upon successful proof verification and nullifier check, the Disbursement Contract is authorized to release the corresponding zakat amount to the claimant's designated address on the Base PoS network.

The circuit design will utilize the Circom domain-specific language for circuit description, with snarkjs for Groth16 proof generation and on-chain verification. The trusted setup ceremony will employ the

Hermez network's Phase 1 Powers of Tau ceremony as the universal SRS, supplemented by a circuit-specific Phase 2 ceremony conducted transparently with documented entropy contributions.

3.4 Phase 3: Prototype Implementation

The prototype implementation targets the Base PoS network as the deployment environment, selected for its EVM compatibility, low transaction fees, and active developer ecosystem. The implementation comprises four main components.

Table 3. Prototype Implementation Components

Component	Description
Circuit Library	Circom circuits implementing the eligibility predicates for the fakir/miskin categories, incorporating Poseidon hash-based commitment schemes, range proof subcircuits for income and asset bounds, and nullifier computation to prevent double-claiming.
DID Registry Contract	Lightweight on-chain DID registry conforming to the did:ethr method specification, storing DID Document hashes and supporting key rotation and deactivation operations compliant with W3C DID Core.
Verifier Contract	On-chain Groth16 proof verification using the Solidity verifier contract generated by snarkjs, integrated with nullifier set management via a Merkle tree data structure for efficient membership proofs.
Claimant Wallet App	Progressive Web Application supporting credential storage, circuit witness generation, proof computation via WebAssembly-compiled prover, and interaction with the smart contract layer via ethers.js.

3.5 Phase 4: Evaluation

The evaluation framework encompasses three dimensions: performance benchmarking, security analysis, and usability assessment. Performance benchmarking will measure proof generation time across hardware profiles representative of target user devices, on-chain verification gas consumption, and end-to-end latency from proof initiation to disbursement execution. These metrics will be compared against alternative privacy approaches to contextualize the protocol's efficiency characteristics.

Security analysis will formally evaluate the protocol against the security properties of completeness, soundness, and zero-knowledge. Additionally, resistance to nullifier-linkage attacks and Sybil attacks through DID-based identity binding will be analyzed. Usability assessment will employ a controlled user study with 30 participants from Indonesian zakat institution staff and prospective mustahik claimants, evaluating the system against the Technology Acceptance Model (TAM) dimensions.

Table 7. Evaluation Framework

Dimension	Metric	Description
Performance	Proof Generation Time	Measured across hardware profiles (smartphone, desktop) representative of target users in Indonesia.
Performance	Gas Consumption	On-chain verification cost per proof; total end-to-end transaction gas on Base PoS.

Performance	End-to-End Latency	Time from proof initiation to disbursement execution confirmation on-chain.
Security	Completeness	A legitimate mustahik can always generate an accepted proof given valid credentials.
Security	Soundness	A non-eligible claimant cannot generate an accepted proof with non-negligible probability.
Security	Zero-Knowledge	The proof reveals no information about private attributes beyond eligibility satisfaction.
Security	Nullifier-Linkage Resistance	Protocol prevents linkage of repeated claims to a single identity across zakat cycles.
Usability	TAM -- Usefulness	Perceived usefulness scored via questionnaire from amil staff participants (n=30).
Usability	TAM -- Ease of Use	Perceived ease of use and Sharia compliance perception from claimant participants.

All experiments will be conducted using Python 3.10 with relevant cryptographic libraries including circomlib, snarkjs, and ethers.js. Smart contracts are deployed to the Base Sepolia testnet before final evaluation on mainnet. Proof generation benchmarks are run on representative hardware configurations including a mid-range Android smartphone (Snapdragon 778G) and a low-specification laptop (Intel Core i5, 8 GB RAM) to reflect realistic deployment conditions in Indonesian zakat institution contexts.

Table 5. Development Tools and Environment

Component	**Tool / Platform**	**Version / Detail**
Programming Language	Python	3.10
ZKP Circuit Language	Circom	2.1.x with circomlib primitives
Proof Generation	snarkjs	Groth16 prover and verifier generation
Smart Contract Language	Solidity	0.8.x -- Verifier and Disbursement contracts
Blockchain Network	Base PoS	EVM-compatible; low-fee mainnet deployment
Wallet Application	Progressive Web App	ethers.js + WebAssembly prover runtime
Experiment Tracking	Weights & Biases	Proof time, gas cost, latency per configuration
Storage	Google Drive / IPFS	VC storage, circuit artifacts, deployment logs

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